Supplementary Material for It's What Gets Delivered That Matters: Communication–Mobility Co-Design over Unreliable Acoustic Links in Cooperative USV–AUV Networks

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***Abstract*—This supplementary material presents the task description, task and system settings, experimental settings, and extended theoretical analysis for the proposed delivery-aware communication–mobility co-design framework. We specify the cooperative underwater sensor-data acquisition mission, vehicle and acoustic-channel models, receiver-local information states, operational routing and resource coordination, and learning/evaluation protocols. We distinguish DATA credited upon AUV reception from ACK-based buffer release and requested routes from locally installed configurations. We further detail timestamped message exchanges, asynchronous configuration acknowledgments, and certificate-based slot leasing, clarifying how limited acoustic airtime constrains executable coordination. Detailed mathematical derivations establish causal state evolution, data and resource conservation, conditional multihop delivery, information-age bounds, disturbance and positioning sensitivity, and consistent cross-timescale reward accounting. The resulting specification links the implemented packet-level task to the assumptions and conclusions of the main manuscript.**

***Index Terms*—Underwater acoustic networks, USV–AUV cooperation, communication–mobility co-design, delivery-aware coordination, information freshness, dynamic topology adaptation.**

I. Task Description

As in the main manuscript, we consider cooperative data acquisition from stationary Internet of Underwater Things (IoUT) sensors by a heterogeneous team of one unmanned surface vehicle (USV) and multiple autonomous underwater vehicles (AUVs). The AUVs navigate among sensors and collect buffered measurements, while the USV supports localization, state exchange, routing coordination, and acoustic-resource management. Because vehicle motion changes link geometry and propagation conditions, the data-acquisition task must be coordinated with the information and network configurations that actually become available after packet delivery.

Each sensor generates timestamped DATA packets and retains them in a finite buffer. An AUV chooses a sensor target, moves into its collection region, and establishes service through DATA_REQUEST and DATA_GRANT messages. DATA reception and buffer release are distinct operations: a packet is credited to the mission after its first valid reception by any AUV, while the sensor removes its retained copy after receiving a corresponding DATA_ACK. Accordingly, multihop AUV–USV routes primarily carry coordination and state information; the evaluated application-payload endpoint is the receiving AUV, instead of the USV or a shore station.

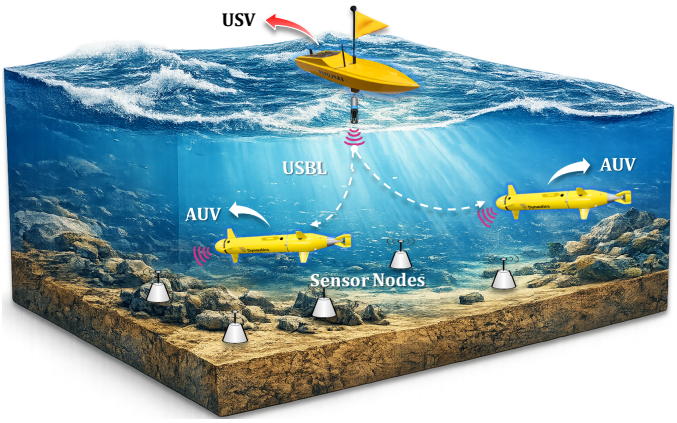


Fig. S1. Cooperative USV–AUV underwater data-acquisition scenario with USV-assisted acoustic localization and AUV collection from stationary seabed sensors.

For a horizon T, let $\mathcal{R}_i(T)$ be the packets received and accepted at AUV i. Distinct sensor–sequence identifiers define the mission-delivery ledger,

$$\begin{aligned} \mathcal{D}(T) &= \bigcup_{i\in\mathcal{A}} \{(j,n) : \mathrm{DATA}_{j,n} \in \mathcal{R}_i(T)\}, \\ C(T) &= |\mathcal{D}(T)|, \qquad R(T) = \sum_{j=1}^{M} R_j(T) \le C(T), \end{aligned} \tag{S1}$$

where $\mathcal{A} = \{1,\dots,N\}$ indexes AUVs and $R_j(T)$ counts distinct packets removed from sensor j following valid acknowledgments. Duplicate receptions do not increase $C(T)$, and the separation between the two counts will be used in the conservation analysis of Section IV.

Beyond successful payload delivery, the mission must account for buffer overflow, communication freshness, propulsion and radio energy, and physical safety. With ΔB_k denoting unique payload bits newly accepted in interval k, ΔO_k the overflowed bits, ΔE_k the energy cost, and $A_i^{\uparrow}(t)$ the age of the latest STATE update accepted at the USV, the main-paper utility is

$$\begin{aligned} U_k = w_B \frac{\Delta B_k}{B_0} - w_O \frac{\Delta O_k}{B_0} - w_E \frac{\Delta E_k}{E_0} \\ - w_A \frac{\sum_{i\in\mathcal{A}} A_i^{\uparrow}(t_{k+1})}{A_0} - w_S \chi_k. \end{aligned} \tag{S2}$$

Here B_0, E_0, and A_0 are positive reference scales; $w_B, w_O, w_E, w_A, w_S \geq 0$ are weights; and χ_k indicates a collision or boundary violation in the interval. In the frozen two-AUV configuration, $(B_0, E_0, A_0) = (2048 \text{ bits}, 100 \text{ J}, 120 \text{ s})$ and $(w_B, w_O, w_E, w_A, w_S) = (5, 2, 0.01, 0.005, 25)$. The motion reward additionally applies potential-based progress shaping with $w_P = 8$, as specified in the main manuscript. The two decision timescales couple this mission objective to AUV motion and to slower USV positioning and topology requests. The following sections specify the resulting physical, networking, and experimental settings before extending the corresponding theory.

II. Task and System Settings

This section provides the physical environment, sensor traffic, acoustic propagation, receiver-local information states, and protocol mechanisms governing the acquisition task. Together, these specifications determine when a selected motion or networking action becomes operational and how its effects are recorded by the packet-level simulator.

A. Environment, Mobility, and Sensor Buffers

The reference two-AUV system comprises one USV, two AUVs, and 30 stationary sensors. The horizontal mission area is $[0, 200]^2$ m^2. The USV operates at depth 0; AUV and sensor depths are fixed at 70 and 100 m. Initial positions and sensor locations are specified by

$$\begin{aligned} L &= 200\,\text{m}, \quad \boldsymbol{q}_j \sim \mathcal{U}([0.08L, 0.92L]^2), \\ \boldsymbol{p}_0(0) &= (0.5L, 0.5L), \\ \boldsymbol{p}_i(0) &= \left(0.30L + 0.40L\frac{i-1}{N-1}, 0.45L\right), \quad N = 2. \end{aligned} \tag{S3}$$

These initial conditions define the simulated geometry; subsequent execution uses each node's locally available position information.

AUV horizontal motion follows the water-relative command plus a spatially and temporally varying current:

$$\dot{\boldsymbol{p}}_i(t) = \boldsymbol{u}_i(t) + \boldsymbol{d}_i(\boldsymbol{p}_i(t), t), \quad \|\boldsymbol{u}_i(t)\| \leq v_{\max}. \tag{S4}$$

The reference commanded speed bounds are $v_{\max} = 2.2$ m/s (AUV) and $v_0 = 1.0$ m/s (USV). The fast decision interval is $\Delta t = 1$ s, with ten physical substeps of 0.1 s. AUV positions use midpoint integration within each substep; the USV advances toward its most recently selected reachable target, instead of teleporting to the next candidate. The base current consists of mean components $(0.20, 0.04)$ m/s, a periodic contribution of nominal amplitude 0.10 m/s and period 120 s, and a vortex with peak speed 0.12 m/s. The actors instead infer their effects from locally measured motion.

Let $d_{\min} = 3\,\text{m}$ and $\Omega = [0, L]^2$. The physical failure indicator and episode endpoint are

$$\begin{aligned} F(t) &= \mathbf{1}\left\{\exists i : \boldsymbol{p}_i(t) \notin \Omega \ \vee \ \min_{i<j} \|\boldsymbol{p}_i(t) - \boldsymbol{p}_j(t)\| < d_{\min}\right\}, \\ T_e &= \min\left\{T_{\max}, \inf\{t : F(t) = 1\}\right\}, \quad T_{\max} = 1200\,\text{s}. \end{aligned} \tag{S5}$$

TABLE SI
Reference two-AUV physical and application configuration.

Parameter	Value
USV / AUV / sensor count	1 / 2 / 30
Horizontal area	200×200 m^2
AUV / sensor depth	70 / 100 m
Episode horizon	1200 s
Fast / slow decisions	1 / 5 s
Physics integration	0.1 s
AUV / USV command speed	2.2 / 1.0 m/s
Sensor collection / safety separation	6 / 3 m
Mean current vector	$(0.20, 0.04)$ m/s
Periodic / vortex current amplitude	0.10 / 0.12 m/s
Displacement noise SD (per axis)	0.025 m
Displacement measurement dropout	0.01
Initial AUV-position noise SD	0.5 m
Range / bearing localization noise SD	0.7 m / 2°
Motion-estimator time constant	5 s
Navigation history capacity	600 s
Sensor initial / maximum buffer	4 / 80 packets
Sensor arrival rate	0.002 packet/s/node

Safety is evaluated at physical-substep resolution, with the learned commands executed directly and early terminations retained in the reported outcomes.

Sensor j begins with four DATA packets and produces $n_{j,k} \sim \text{Poisson}(\lambda \Delta t)$ new packets each fast interval, where $\lambda = 0.002$ packet/s. The buffer capacity is $Q_{\max} = 80$ packets. Arrivals are admitted before service; if $q_{j,k}$ is occupancy before arrivals and $r^{\text{ack}}_{j,k}$ denotes distinct packets released on valid acknowledgments, then

$$o_{j,k} = [q_{j,k} + n_{j,k} - Q_{\max}]^+, \tag{S6}$$

$$q_{j,k+1} = q_{j,k} + n_{j,k} - o_{j,k} - r^{\text{ack}}_{j,k}. \tag{S7}$$

Entering the 6-m horizontal collection disk initiates a service opportunity, which then requires scheduled and delivered DATA_REQUEST, DATA_GRANT, DATA, and DATA_ACK exchanges. The data-range criterion applies at the relevant transmission/reception event, while the physical error calculation uses three-dimensional range. AUVs maintain queue beliefs from known initial loads, arrival-rate priors, and received reports or grants rather than reading sensor queues directly.

For $Q_0 = 4$, $M = 30$, and $T_{\max} = 1200$ s, the offered-data expectation is

$$\mathbb{E}[N_{\text{offered}}] = M(Q_0 + \lambda T_{\max}) = 30(4 + 0.002 \times 1200) = 192. \tag{S8}$$

The offered load provides a supply-side reference; actual delivery further depends on collection geometry, handshakes, slot availability, acoustic losses, and mission duration.

B. Acoustic Communication, Slot Access, and Energy

The sensor-service rule specifies when a packet exchange may be initiated. We next quantify the propagation, airtime, and energy costs governing successful packet exchange.

The reference acoustic bitrate is $R = 16$ kbit/s over nominal bandwidth $W = 8$ kHz, and the propagation speed is $c_s = 1500$ m/s. Transmitted DATA carries $B_p = 2048$ payload bits plus a 256-bit header; nominal STATE and other control packets occupy 1536 and 512 bits, respectively. The reference SNR at 100 m is 18 dB, with logarithmic spreading coefficient 15 dB

TABLE SII
REFERENCE ACOUSTIC, ROUTING, AND SERVICE PROTOCOL PARAMETERS.

Parameter	Value
Bitrate / bandwidth	16 kb/s / 8 kHz
Sound speed / maximum link range	1500 / 350 m
SNR at 100 m / error floor	18 dB / 0.015
Bad-link packet-error floor	0.70
Slot / frame duration	0.6 / 36 s
Home-slot frame length	60 slots
Propagation guard / processing delay	0.04 / 0.02 s
STATE generation / TTL	5 / 90 s
HELLO period / report TTL	20 / 90 s
CONFIG timeout / route expiry	100 / 300 s
CONFIG retries / switch dwell	3 / 15 s
Service ACK timeout / grant TTL	80 / 100 s
Sensor-target min / max dwell	30 / 300 s
Transmission / reception / listening	5 / 1 / 0.05 W

and range-dependent absorption 4 dB/km. The implemented packet-error approximation uses

$$\Gamma_{\mathrm{dB}}(D) = 18 - 15\log_{10}\frac{\max(D, 1\ \mathrm{m})}{100\ \mathrm{m}} - 0.004(D - 100\ \mathrm{m}), \tag{S9}$$

$$p_b(D) = \tfrac{1}{2}\,\mathrm{erfc}\left(\sqrt{10^{\Gamma_{\mathrm{dB}}(D)/10} W/R}\right), \tag{S10}$$

$$p_e(D, b) = 1 - (1 - p_0)[1 - p_b(D)]^b, \quad p_0 = 0.015. \tag{S11}$$

A two-state burst component can raise the packet-error probability to at least 0.70 in its bad state; the native bad-state transition probabilities per simulator update are 0.003 (good-to-bad) and 0.2 (bad-to-good). Range cutoffs, explicitly injected outages, and DATA collection-range checks may independently prevent delivery. Queueing precedes transmission, and a packet with b_m total bits has nominal arrival time

$$\begin{aligned} t_m^{\mathrm{arr}} &= t_m^{\mathrm{tx}} + \frac{b_m}{R} + \frac{D_{vw}(t_m^{\mathrm{tx}})}{c_s} + t_{\mathrm{proc}}, \\ \zeta_{m,w}(t) &= \mathbf{1}\{t_m^{\mathrm{arr}} \le t\}\xi_m \vartheta_{m,w}. \end{aligned} \tag{S12}$$

Here ξ_m is physical reception and $\vartheta_{m,w}$ is the application/protocol acceptance indicator. Only $\zeta_{m,w}(t) = 1$ changes the receiver's information state.

The reference TDMA slot is $\delta = 0.6$ s. Ten groups allocate one home slot to each cooperative node and three selected sensors per group, yielding

$$S = 10(1 + N + 3) = 60 \quad (N = 2), \qquad T_{\mathrm{fr}} = S\delta = 36\,\mathrm{s}. \tag{S13}$$

The 0.6-s slot accounts for maximum declared packet serialization, geometric propagation, processing, and a guard allowance. The network implementation checks that the configured slot is sufficiently long. The conflict-free access model assumes synchronized slots, compliant transmissions, and the declared packet-size bound.

The modeled propulsion and total episode energy obey

$$\begin{aligned} P_i^{\mathrm{prop}}(t) &= 12 + 2\|\boldsymbol{u}_i(t)\|^2 + \|\boldsymbol{u}_i(t)\|^3, \quad i \in \mathcal{A}, \\ P_0^{\mathrm{prop}}(t) &= 15 + 2\|\boldsymbol{u}_0(t)\|^2, \\ E_e &= \int_0^{T_e} \left(\sum_{i=0}^{N} P_i^{\mathrm{prop}}(t) + \sum_{v \in \mathcal{N}} P_v^{\mathrm{radio}}(t)\right) dt. \end{aligned} \tag{S14}$$

Power is in watts and E_e in joules; the radio term includes transmission, reception, and listening even for failed messages. Since termination changes both T_e and credited C_e, total energy E_e and energy per accepted packet E_e/C_e are distinct metrics (the latter is undefined if $C_e = 0$).

C. Causal Information, Estimation, and Motion Decisions

With packet reception and channel access specified, we next construct the information available to each agent. Controls use receiver-local evidence, and source-time localization corrections remain separate from displacement-based current estimates.

For any executing node v, let $\mathcal{H}_v(t)$ collect its initial knowledge, local measurements/commands, protocol timers, and accepted packet set $\mathcal{R}_v(t)$. Its action is adapted to that history:

$$\begin{aligned} \mathcal{R}_v(t) &= \{m : \zeta_{m,v}(t) = 1\}, \\ \boldsymbol{o}_v(t) &= \phi_v(\mathcal{H}_v(t)), \\ a_v(t) &\sim \pi_v(\cdot \mid \boldsymbol{o}_v(t)). \end{aligned} \tag{S15}$$

An AUV thus sees only local navigation, motion evidence, installed routes, received service grants and peer reports; the USV sees receiver-held STATE/link reports, acknowledged configurations, and service evidence. Physical truth drives simulator transitions and offline metrics, whereas executing agents use the states constructed from local histories.

For the evaluated two-AUV implementation, the AUV observation has 41 entries and the USV topology observation has 90. The 41-dimensional AUV schema groups into: (i) two position and two target displacement components; (ii) six executed/measured/disturbance-motion components; (iii) measurement validity, initialization and measurement age features; (iv) TDMA and sensor-slot phases; (v) target queue, distance, and dwell features; (vi) localization age and uncertainty; (vii) route, COORD, grant and pending-request validity/age fields; and (viii) relative USV and peer-report geometry. The observation features follow a fixed ordering that is maintained consistently during training and evaluation. Root-side features include timestamped AUV measurements, the directed report-quality/age matrix, active and pending parent vectors, acknowledgments, and task/service summaries.

The disturbance estimator uses an independently measured displacement $\Delta\boldsymbol{p}_i^m$, not the difference of successively corrected absolute positions. Over a valid interval of duration h,

$$\boldsymbol{z}_i = \frac{\Delta\boldsymbol{p}_i^m - \int_{t-h}^{t} \boldsymbol{u}_i(s)\,ds}{h}, \tag{S16}$$

$$\widehat{\boldsymbol{d}}_i^{+} = \widehat{\boldsymbol{d}}_i^{-} + (1 - e^{-h/\kappa})(\boldsymbol{z}_i - \widehat{\boldsymbol{d}}_i^{-}), \quad \kappa = 5\ \mathrm{s}. \tag{S17}$$

With invalid displacement sensing (native dropout probability 0.01), the estimator retains its value and increases the age of its evidence; the navigation filter may propagate using previously available commands but cannot reclassify the prediction as a new measurement. A displacement measurement has nominal independent per-axis noise SD 0.025 m; the initial local-position estimate has SD 0.5 m.

An acoustic fix reports the AUV position at the measurement source time, not at reception. The simulator samples a range/direction-based position observation using nominal

0.7-m range and 2° bearing noise. When its reply reaches the AUV, the local filter reconstructs the position estimate at the source time using retained navigation increments; a scalar nominal covariance then determines a source-time innovation gain. The frozen planning/fusion option uses a nominal fix SD of 3.0 m, which is an algorithmic covariance choice rather than an assertion that all geometry-dependent errors have this SD. Duplicate, future-dated, out-of-order, or history-expired fixes are rejected. Consequently, delayed corrections are incorporated into navigation without contaminating the disturbance residual.

Each AUV actor outputs $\boldsymbol{a}_i = (a_i^v, a_i^\theta) \in [-1,1]^2$. With the locally observed target bearing β_i, speed and heading are decoded as

$$v_i = \frac{a_i^v + 1}{2} v_{\max}, \quad \theta_i = \beta_i + \pi a_i^\theta,$$
$$\boldsymbol{u}_i = v_i(\cos\theta_i, \sin\theta_i)^\mathsf{T}. \tag{S18}$$

This target-relative action encoding spans the velocity disk; a separate fixed-time disturbance compensation or safety override is not appended to the TD3 command. The USV evaluates a finite reachable candidate set consisting of its current position and up to 32 polar candidates (two rings of 16 angles). The candidate score $J_0(\boldsymbol{x})$ combines FIM geometry, movement cost, confirmed direct-route obligations, and aged received link evidence:

$$\widehat{q}_{ab}(t) = q_{ab}^{\mathrm{rep}} e^{-(t-t_{ab}^{\mathrm{rep}})/T_{\mathrm{rep}}} \times \mathbf{1}\{0 \le t - t_{ab}^{\mathrm{rep}} \le T_{\mathrm{rep}}\}, \tag{S19}$$
$$\rho_i = \widehat{q}_{i0}\widehat{q}_{0i}.$$

The USV chooses $\boldsymbol{x}_0^\star = \arg\min_{\boldsymbol{x}\in\mathcal{P}_0(t)} J_0(\boldsymbol{x})$ and uses a bounded velocity-persistence predictor with $\tau_v = 3$ s in the frozen options. In the score J_0, the FIM term has unit weight, and the direct-route and movement coefficients are $(\eta_R, \eta_M) = (0.2, 0.1)$. The predictor provides a bounded extrapolation from the latest delivered state and command.

The reference topology catalogue has rooted parent vectors $(0,0)$, $(0,1)$, and $(2,0)$, plus a KEEP/repair action. The selected Double-DQN action is filtered by a received-evidence admissibility mask. For example, a relay request depends on adequately recent and appropriate bidirectional reports, pending protocol state, and minimum dwell. The admissibility mask thus evaluates the evidence supporting a request, while route installation and physical delivery are resolved subsequently by the protocol.

D. Operational Routing, Slot Leasing, and Task Admission

The selected motion and topology decisions are next translated into executable network actions. Route installation, transmission rights, and service grants are therefore updated through received protocol messages.

The protocol separates a requested route version ν, a locally installed version, and the USV's confirmed version. Let $\mathcal{C}_v(t)$ be node v's route records, $\mathcal{A}_\nu^{\mathrm{req}}$ the acknowledgments required by the transaction, and $\mathcal{A}_\nu^{\mathrm{ack}}(t)$ the acknowledgments actually received at the USV:

$$I_v^\nu(t) = \mathbf{1}\{(\nu, \boldsymbol{b}^\nu, e^\nu) \in \mathcal{C}_v(t),\ t < e^\nu\},$$
$$I_0^{\nu,\mathrm{conf}}(t) = \mathbf{1}\{\mathcal{A}_\nu^{\mathrm{req}} \subseteq \mathcal{A}_\nu^{\mathrm{ack}}(t),\ t < e^\nu\}. \tag{S20}$$

A received CONFIG may activate a local route before CONFIG_ACK returns, so $I_v^\nu = 1$ need not imply $I_0^{\nu,\mathrm{conf}} = 1$.

The resource mechanism begins with a preinstalled fixed-owner calendar. For a temporary transfer from donor d to borrower b, the donor first stores a silence commitment after a received PREPARE; only a subsequently delivered donor acknowledgment permits COMMIT. The borrower activates rights only on receiving a valid authorization. The USV can make bounded sub-delegations to sensors through a charged SUBCOMMIT from an already committed parent window. The USV records its own silence before issuing the child certificate. If a certificate is lost, the associated silence commitment remains effective until its prescribed expiration, potentially leaving the interval idle. All coordination traffic is counted as packet transmissions, and resource requests are based on locally received demand evidence.

For sensor j, the AUV uses queue belief $\widehat{q}_{ij}$, an arrival prior λ_j, and its known home-slot calendar to estimate service time. The resulting admission score and target choice are

$$T_{ij}^{\mathrm{tr}} = \frac{[\|\widehat{\boldsymbol{p}}_i - \boldsymbol{q}_j\| - r_c/2]^+}{\bar{v}},$$
$$\widehat{T}_{ij} = T_{ij}^{\mathrm{tr}} + T^{\mathrm{hs}} + T_{ij}^{\mathrm{sv}},$$
$$\mathcal{U}_{ij}(t) = \frac{B_p \widehat{n}_{ij}}{\max\{\Delta t, \widehat{T}_{ij}\}}, \qquad g_i^\star = \arg\max_{j\in\mathcal{T}_i^{\mathrm{adm}}} \mathcal{U}_{ij}(t). \tag{S21}$$

Here $\widehat{n}_{ij}$ is the predicted deliverable packet count and $\mathcal{T}_i^{\mathrm{adm}}$ the received-evidence-based admissible set. The visit score incorporates service opportunities known to the selecting AUV.

To reduce duplicate service effort, newly chosen targets near a received and still fresh peer target may be excluded within nominal radius $2r_c + d_{\min}$. Current legitimate visits and the AUV's own unexpired received grant are preserved. A losing AUV already within the winner's service region may prefer targets whose nominal direction satisfies $(\widehat{\boldsymbol{p}}_i - \boldsymbol{c})^\mathsf{T}(\boldsymbol{q}_j - \widehat{\boldsymbol{p}}_i) \ge 0$. This preference discourages inward target switching; physical collision checks remain part of the executed-motion evaluation. When no admissible alternative exists, the previous task is retained.

III. Experimental Settings

Having established the task and operational rules, we next describe the learning configuration, comparison systems, disturbance schedules, and statistical measures used in the main paper. All evaluations employ the same payload-credit definition and receiver-local execution semantics introduced above.

A. Learning Configuration and Event-Driven Evaluation

These interacting mechanisms are trained on executed packet-level outcomes and evaluated using frozen checkpoints. The training configuration and simulator event order below connect the analytical protocol to the experimental observations.

In the reference two-AUV setting, each AUV uses independent TD3 [2] actor/twin-critic modules with two 128-unit hidden layers, batch size 64, learning rate 3×10^{-4}, replay capacity 10^5, discount $\gamma = 0.995$, target soft-update

TABLE SIII
OPERATIONAL MESSAGE CLASSES AND ACCEPTANCE CONSEQUENCES. MESSAGES ARE SERIALIZED THROUGH THE ACOUSTIC SCHEDULER; THE TABLE DOES NOT IMPLY DETERMINISTIC DELIVERY.

Message class	Purpose and relevant fields	Effect only after valid reception / acknowledgment
HELLO / HELLO_ACK / REPORT	Link discovery, directed-quality observations and timestamps	Update a receiver's report cache; do not reveal future or undelivered link states.
STATE / COORD	AUV state, task intention, locally held peer/route evidence	Update USV/AUV information caches only if timestamp and validity rules admit the message; relaying preserves source age.
CONFIG / CONFIG_ACK	Requested parent vector, version, lease expiration, acknowledgments	CONFIG installs a local route; CONFIG_ACK received at the USV supports confirmation, which may lag installation.
PREPARE / donor acknowledgment / COMMIT	Donor silence promise, borrower, slot-window certificate	Donor relinquishes promised windows before acknowledgement; borrower activates only on receipt of a valid commitment.
SUBCOMMIT	Child authorization within an already owned USV window	Receiving sensor obtains nonoverlapping temporary rights; the parent must remain silent for that window.
DATA_REQUEST / DATA_GRANT	Service request, sensor buffer report, grant lifetime	Service availability changes only after received request and grant; a remote grant is not assumed from root intentions.
DATA / DATA_ACK / service-END	Sensor sequence, distinct payload, release acknowledgment, generation-aware end notice	Distinct DATA is credited at the AUV; ACK releases the sensor buffer; a delivered END clears the remote service occupancy.
LOCALIZATION_PING / LOCALIZATION / LOCALIZATION_FIX	Charged localization interrogation, measured position, source timestamp	A time-stamped fix is fused at its source time, only after successful reply delivery and validity checking.

coefficient 0.005, and delayed actor update period 2. Target action noise has SD 0.2 with clipping ± 0.5. The actor objective is $\mathcal{L}_\mu = -\mathbb{E}[Q_{\theta_1}(\boldsymbol{o}, \mu_\omega(\boldsymbol{o}))] + \lambda_n \mathcal{L}_{\text{aux}}$, with λ_n decaying from 0.2 to 0.02 over 10^5 critic updates and auxiliary command penalty $\lambda_u = 0.02$. During deployment, the actor continues to use the command decoding in (S18).

The slow learner uses masked Double DQN [3] with a realized d_ℓ-step target

$$\begin{aligned} R_\ell &= \sum_{j=0}^{d_\ell - 1} \gamma^j r^R_{k_\ell + j}, \\ y^R_\ell &= R_\ell + (1 - \chi_\ell)\gamma^{d_\ell} Q_{\bar{\psi}}(\boldsymbol{z}'_\ell, a^\star_\ell), \\ a^\star_\ell &= \arg \max_{a: m'_\ell(a)=1} Q_\psi(\boldsymbol{z}'_\ell, a). \end{aligned} \tag{S22}$$

Its additional reward terms penalize normalized upward STATE age, confirmed downward COORD age, and new configuration requests. The implemented coefficients are $(c_r, c_u, c_d, c_c) = (0.2, 0.08, 0.04, 0.1)$, which impose explicit freshness and reconfiguration preferences on topology learning.

The principal Full and USDE-only policies are evaluated at 450,000 environment steps after continuing from their respective 100,000-step training states with replay and optimizer states preserved. The two methods use comparable TD3 architectures and nominal training hyperparameters, while their networking and task-coordination capabilities differ as specified below. The adapted FIM+RL and communication-aware references retain their method-specific training procedures. For the three- and four-AUV extensions, policies are separately trained for 450,000 steps using fleet-specific observation and communication interfaces. Evaluation freezes the selected checkpoints and disables exploration; the scenario-level intervals described below summarize variation across paired evaluation conditions rather than treating packet transmissions or AUV-time observations as independent samples.

Algorithm 1 summarizes the resulting packet-level execution loop. During evaluation, policies are frozen; during training, eligible transitions additionally trigger scheduled TD3 and Double-DQN updates. Simulator transitions and rewards use the physical environment, while policy inputs follow the receiver-local states defined above.

Algorithm 1 Receiver-local packet-level episode execution

Require: Frozen policies, valid local initial states, scenario seed

for $k = 0, 1, \ldots, K_{\text{end}} - 1$ **do**
  if $k \bmod K = 0$ **then**
    Build USV evidence state from received reports.
    Select the admissible topology request; enqueue CONFIG as needed.
    Evaluate the reachable USV targets, and publish charged COORD.
  end if
  Each AUV decodes TD3 action from its local state.
  Generate sensor arrivals; enqueue due STATE, service and localization packets.
  for each 0.1-s physical substep **do**
    Advance USV/AUV motion and test physical safety.
    Advance scheduled acoustic slots and event queue.
    Apply the received packets to the receiver-local protocol state.
    Enqueue the permitted DATA, replies, and acknowledgments.
  end for
  Update independent displacement sensing and USDE.
  Fuse newly delivered, valid delayed localization fixes.
  Update local tasks; record executed mission metrics.
  if collision, boundary violation, or time limit **then**
    Preserve termination cause; stop the episode.
  end if
end for

B. Reference Methods and Evaluation Coverage

The evaluation compares the complete proposed system (Full) with USDE-only, adapted FIM+RL, and adapted communication-aware (CA) references under common physical motion, acoustic-channel, traffic, termination, and delivery-accounting rules. Full combines disturbance-aware AUV motion, received-evidence-dependent topology and USV decisions,

versioned route installation, dynamic slot leasing, and delivered-service-based task selection. USDE-only preserves the disturbance estimator and baseline collection procedure but uses static direct routes without the added network-resource and task-coordination mechanisms. The two adapted references preserve their method-specific planning and channel-access designs. Accordingly, these tests compare integrated system designs, while the deployment interventions isolate individual coordination mechanisms under frozen Full policy weights.

Across the study, the main manuscript reports $N_{\mathrm{eval}} = 5760$ packet-level evaluation episodes. The seven groups of experiments are summarized in Table SIV, with 30 predefined scenarios per method–condition cell. Within a comparison, the same scenario definitions and termination rules are applied to each system.

The mechanism interventions separately suppress new slot leases, topology requests, received peer intent, or service-admission/yielding behavior, while preserving Full's trained policy weights. These interventions evaluate the effect of execution-level changes without redefining the common physical mission. The next subsection specifies the corresponding environmental disturbances and fleet-specific calendars.

C. Fault Injection and Fleet-Specific Configurations

Having defined what each reference system can do, we next specify the environmental and network interventions under which those systems are compared. Disturbances are preset; position-dependent exposure may still vary with the executed trajectories.

The reference environment already includes current disturbances, random and bursty acoustic errors, and a bidirectional USV–AUV1 outage during $[200, 350)$ s. Additional experiment-specific impairments are applied without removing those baseline disturbances. The evaluation conditions are summarized in Table SV; the same externally specified scenario is paired across competing systems, but realized position-dependent link exposure may differ because each policy moves differently.

For stress level $\ell \in \{1, 2, 3, 4\}$, define the imposed parameter vector $\boldsymbol{\upsilon}_\ell = (u_\ell, p_\ell, b_\ell, T_{1\ell}, T_{2\ell}, T_{r\ell})$, with ordered values

$$\begin{aligned} \boldsymbol{u}_{\mathrm{pk}} &= (0.10, 0.25, 0.50, 0.80)\ \mathrm{m/s}, \\ \boldsymbol{p}_{\mathrm{bad}} &= (0.25, 0.50, 0.70, 0.90), \quad \boldsymbol{b} = (2, 5, 10, 20)\ \mathrm{m}, \\ \boldsymbol{T}_1 &= (20, 50, 100, 200)\ \mathrm{s}, \quad \boldsymbol{T}_2 = (5, 10, 20, 40)\ \mathrm{s}, \\ \boldsymbol{T}_r &= (5, 15, 40, 80)\ \mathrm{s}. \end{aligned} \tag{S23}$$

These respectively specify peak added currents, bad-state erasure probabilities, localization biases, two root-link interruption durations, and AUV-radio interruption durations. Sensor DATA continues to require direct sensor-to-AUV reception. A radio outage disables communication for the affected AUV while its physical body continues moving; it is not a removal of the vehicle or its safety checks.

The multi-AUV extension uses 30 sensors in the same nominal area, but its calendar profile is explicitly adapted. The fleet-dependent calendar $\boldsymbol{\kappa}_N = (S_N, \delta_N, T_{\mathrm{fr},N})$ is

$$\boldsymbol{\kappa}_N = \begin{cases} (60, 0.6, 36), & N = 2, \\ (50, 0.5, 25), & N = 3, \\ (55, 0.5625, 30.9375), & N = 4, \end{cases} \tag{S24}$$

where δ_N and $T_{\mathrm{fr},N}$ are in seconds. The three-/four-AUV profiles use five groups and six sensors per group, giving $5(4+6)$ and $5(5+6)$ slots respectively. Its root resource pool and admissible observation/topology interfaces are also adapted. Within a given fleet size, the corresponding Full and matched-USDE systems share the fleet-specific rules. The fleet-size comparison thus reflects the joint effects of team size and its associated communication profile. Results characterize the implemented three- and four-AUV configurations.

D. Metrics, Pairing, and Statistical Reporting

To interpret these evaluations, credited payload, observed AoI, mission survival, and uncertainty must use compatible denominators. The following definitions also separate paired scenario variability from training-seed variability.

For an episode e, let C_e be distinct sensor identifiers accepted by any AUV, $T_e \le T_{\max}$ be its actual duration, and N_e the number of AUVs. The fixed-budget goodput is

$$G_{\mathrm{fixed,e}} = \frac{B_p C_e}{T_{\max}}, \qquad G_{\mathrm{active,e}} = \frac{B_p C_e}{T_e}. \tag{S25}$$

They coincide only for full-length episodes. The pooled STATE age weights actually executed AUV–time rather than equally weighting episode means:

$$\overline{A}_{\mathrm{pool}} = \frac{\sum_e \int_0^{T_e} \sum_{i=1}^{N_e} A_{i,e}^{\uparrow}(t)\, dt}{\sum_e N_e T_e}. \tag{S26}$$

A tabulated quantity of the form $x \pm s$ is the episode mean and sample SD, while $x\ (s)$ denotes a pooled value with the SD of a corresponding episode-level statistic in parentheses. Pooled proportions and percentiles are computed using their corresponding pooled denominators. DATA non-delivery uses failed resolved hop attempts divided by all resolved DATA hop attempts, including hard faults and out-of-range attempts; it is not the uncoded physical-channel bit- or packet-error probability. Successful transmission attempts still may not yield new unique payload if the packet duplicates one already accepted.

For outage and stress analyses, window-limited information availability is also useful. Let $W = [t_a, t_b]$ be a planned window, and define $V_{i,e}(t) = \mathbf{1}\{t \le T_e,\ A_{i,e}^{\uparrow}(t) \le D\}$. The planned-budget availability ratio is

$$\mathcal{V}_D(W) = \frac{\sum_e \sum_i \int_{t_a}^{t_b} V_{i,e}(t)\, dt}{(t_b - t_a) \sum_e N_e}. \tag{S27}$$

The indicator vanishes after episode termination, so this planned-window metric captures both age availability and mission survival. It is reported alongside the survivor-weighted pooled AoI. Physical completion and reasons for termination are always reported alongside both measures. In the combined-stress screen, the available-information threshold is 60 s over

TABLE SIV
MAIN-MANUSCRIPT EXPERIMENTAL COVERAGE UNDER PAIRED, PACKET-LEVEL EVALUATION. EACH METHOD–CONDITION CELL CONTAINS 30 SCENARIOS.

Evaluation suite	Purpose and coverage	Episodes
Primary system comparison	Four integrated systems under the common two-AUV reference scenario	120
Acoustic-link disturbances	Eleven link-loss, outage, SNR, and partition conditions	1320
Additional ocean currents	Thirteen reference and added-current conditions	1560
Slow-decision scheduling	Three scenarios with eight scheduling modes or reference configurations	720
Mechanism interventions	Three scenarios with six reference or fixed-policy deployment configurations	540
Fleet-size extensions	Reference and outage conditions for two-, three-, and four-AUV context and matched systems	300
Combined stress and timing	Joint-disturbance and timing conditions for the specified system comparisons	1200
Total		**5760**

TABLE SV
EXPERIMENTAL DISTURBANCE SETTINGS AND FLEET CONFIGURATIONS. INTERVENTION TIMES AND MAGNITUDES ARE DEFINED BEFORE EVALUATION.

Suite	Conditions	Applied settings
Acoustic-link faults	Reference; a single link lost for 120 or 300 s; alternating links; complete root partition; extra 10%/30% erasures; burst loss; AUV radio outage; local $-6/-12$ dB SNR	Extra events begin at 500 s; the 300-s single-link interruption, partition and radio outage end at 800 s. Alternation changes the affected direct link at 650 s. Local SNR impairments use a fixed 55-m-radius disk centered at (100,100) m and operate on actual transmitter/receiver geometry.
Additional currents	Reference plus constant, periodic, cross-current gust and correlated random flows at 0.10, 0.25 and 0.40 m/s	Additions act during 500–800 s except the 500–560-s gust. Transitions are smoothed over 5 s. Physical flow changes trajectories and subsequent communications; a policy does not directly observe the injected ground-truth current.
Slow-clock coordination	USV positioning only; topology adaptation only; motion-first; topology-first; simultaneous; plus three reference schedules	Five matched runtime schedules use a 30-s common initialization and one update of each enabled module every 10 s. Native 5-s Full is an additional, more-frequently updated anchor, not an equal-budget direct competitor.
Mechanism interventions	Full; disable admission/yielding, peer intent, new leases or new route requests; trained USDE-only reference	Reference and separate AUV1/AUV2 direct-root outages during 400–800 s; intervention variants use frozen Full policies and matched evaluation seeds.
Combined stress	Reference, four joint levels and five moderate single-factor controls	Currents, correlated erasures and localization bias act during 400–800 s; root-link and AUV-radio interruptions begin at 500 s. Weak/moderate/severe/extreme current peaks: 0.10/0.25/0.50/0.80 m/s.
Fleet extension	Native 2-AUV context; 3/4 AUV Full and matched USDE	Reference and 400–800-s root-link outage. Three-/four-AUV configurations have fleet-specific training, radio calendars and interface dimensions; the two-AUV point is historical context rather than a controlled single-factor scale manipulation.

the planned 400-s stress window; this is an empirical screen, not the 90-s STATE TTL or a theorem-derived safety boundary.

Each published method–condition cell uses $n = 30$ pre-assigned paired scenarios, including early terminations. For paired payload outcomes C_e^F and C_e^B, the experimental unit and paired difference are

$$\Delta_e = C_e^F - C_e^B, \quad \widehat{\Delta} = \frac{1}{n}\sum_{e=1}^{n}\Delta_e, \qquad s_\Delta^2 = \frac{1}{n-1}\sum_{e=1}^{n}(\Delta_e - \widehat{\Delta})^2. \tag{S28}$$

Bootstrap resampling operates on complete paired scenarios, preserving dependencies among their AUV-time observations and packet attempts. Ratio and pooled-AoI statistics are recomputed within each bootstrap replicate from their numerator and denominator. The primary comparisons use 20,000 paired resamples; the fleet-comparison analysis uses 40,000 resamples with an eight-comparison adjustment. Other protocol analyses use their documented pointwise resample settings, which should not be interpreted as globally multiplicity-adjusted evidence.

The primary evaluation uses scenario seeds 9,300,000–9,300,029; separate predetermined ranges are assigned to additional link failures (9,620,000–9,620,029), current disturbances (9,630,000–9,630,029), slow-clock scheduling (9,650,000–9,650,029), mechanism interventions (9,660,000–9,660,029), and combined stress (9,690,000–9,690,029). Fleet-extension experiments use separate reference and outage scenarios. The 30 paired scenarios characterize variability under frozen policies, whereas variability across training runs forms a separate statistical level. The reported statistics are reconstructed from complete episode outcomes and timestamped packet-event records under the fixed scenario definitions.

IV. EXTENDED THEORETICAL ANALYSIS

The analysis proceeds from receiver-local causality and packet conservation to exclusive acoustic-resource authorization and conditional multihop delivery. It then connects the availability of accepted messages to information-age bounds, disturbance and motion-estimation uncertainty, USV positioning sensitivity, and communication-resource accounting. Deterministic protocol invariants and conditional stochastic performance bounds are stated with their respective assumptions.

A. Local Causality under Asynchronous Events

Let $\mathscr{F}_v(t)$ denote the information generated by node v's prior local measurements, commands, received-and-accepted packet records, timers, installed routes, resource certificates, and local randomized decisions. The simulator's physical and network truth can depend on a larger filtration $\mathscr{G}(t)$, but is not available to node v unless encoded by a permitted local observation or delivered packet.

Proposition S.1 (Indistinguishability of remote execution histories)**.** *Consider two admissible executions initialized with identical local state and policy randomization at node v. If they induce identical time-ordered local events and accepted packet contents through time t, then all state variables updated at v—including its installed route, valid leases, local estimates, and selected actions—are identical through t, even if remote packet losses, requests, or true positions differ.*

Proof: Index the local state-changing events of node v by n, and let $\boldsymbol{X}_{v,n}^{(r)}$ be its post-event state in execution $r \in \{1,2\}$. Denote the event contents, including accepted packets and sensor readings, by $\mathcal{E}_{v,n}^{(r)}$, and the coupled local random input by $\omega_{v,n}^{(r)}$. The receiver-local transition takes the form

$$\begin{aligned} \boldsymbol{X}_{v,n+1}^{(r)} &= \mathfrak{F}_v\big(\boldsymbol{X}_{v,n}^{(r)}, \mathcal{E}_{v,n+1}^{(r)}, \omega_{v,n+1}^{(r)}\big), \\ \boldsymbol{X}_{v,0}^{(1)} &= \boldsymbol{X}_{v,0}^{(2)}, \quad (\mathcal{E}_{v,n}^{(1)}, \omega_{v,n}^{(1)}) = (\mathcal{E}_{v,n}^{(2)}, \omega_{v,n}^{(2)}). \end{aligned} \tag{S29}$$

For the first event, equality of the arguments of $\mathfrak{F}_v$ implies $\boldsymbol{X}_{v,1}^{(1)} = \boldsymbol{X}_{v,1}^{(2)}$. Suppose the post-event states agree through event n. Then

$$\begin{aligned} \boldsymbol{X}_{v,n}^{(1)} &= \boldsymbol{X}_{v,n}^{(2)} \\ \implies \quad & \mathfrak{F}_v(\boldsymbol{X}_{v,n}^{(1)}, \mathcal{E}_{v,n+1}, \omega_{v,n+1}) \\ & = \mathfrak{F}_v(\boldsymbol{X}_{v,n}^{(2)}, \mathcal{E}_{v,n+1}, \omega_{v,n+1}) \\ \implies \quad & \boldsymbol{X}_{v,n+1}^{(1)} = \boldsymbol{X}_{v,n+1}^{(2)}. \end{aligned} \tag{S30}$$

Between consecutive event times $t_n < t < t_{n+1}$, both executions use the same locally determined clock and held-command evolution, say $\boldsymbol{X}_v^{(r)}(t) = \Psi_v(\boldsymbol{X}_{v,n}^{(r)}, t - t_n)$. Thus $\boldsymbol{X}_v^{(1)}(t) = \boldsymbol{X}_v^{(2)}(t)$ for all t in the interval. Induction over the ordered event sequence proves the claim; an unreceived remote event does not enter $\mathcal{E}_{v,n}$. ∎

This establishes the execution-time causality of receiver-local state updates. In particular, global DATA accounting remains an offline evaluation quantity, while each agent acts on its own event history. We next derive the corresponding conservation laws for sensor payload.

B. Sensor Inventory and Unique-Payload Conservation

Because actions can depend only on accepted local evidence, a subsequent performance claim must also respect the distinction between receiving a DATA packet and removing its copy from the sensor buffer.

Let $Q_{j,0}$ be sensor j's initial packets, $N_j(T)$ its admitted-plus-overflow arrival count, $O_j(T)$ its cumulative overflow, $R_j(T)$ its cumulative ACK-confirmed release count, and $Q_j(T)$ its ending occupancy. Each quantity counts distinct sensor packet instances, not transmissions.

Proposition S.2 (Inventory and delivery accounting)**.** *For every realized episode, under the queue update in* (S7)*,*

$$Q_{j,0} + N_j(T) = Q_j(T) + O_j(T) + R_j(T). \tag{S31}$$

If a valid ACK only releases an instance previously accepted at an AUV, then $\sum_j R_j(T) \le C(T)$, where $C(T)$ is the number of unique DATA instances accepted by any AUV by time T. Furthermore,

$$C(T) \le \sum_j \big(Q_{j,0} + N_j(T) - O_j(T)\big). \tag{S32}$$

Proof: For a realized episode with K_e executed queue updates, write $N_j = \sum_{k=0}^{K_e-1} n_{j,k}$, $O_j = \sum_{k=0}^{K_e-1} o_{j,k}$, and $R_j = \sum_{k=0}^{K_e-1} r_{j,k}^{\mathrm{ack}}$. Summing (S7) gives

$$\begin{aligned} \sum_{k=0}^{K_e-1} (q_{j,k+1} - q_{j,k}) &= \sum_{k=0}^{K_e-1} (n_{j,k} - o_{j,k} - r_{j,k}^{\mathrm{ack}}) \\ q_{j,K_e} - Q_{j,0} &= N_j - O_j - R_j, \\ Q_{j,0} + N_j &= q_{j,K_e} + O_j + R_j. \end{aligned} \tag{S33}$$

To relate buffer release to delivered payload, let $\mathcal{I}_j$ be the set of distinct generated DATA keys, $\mathcal{O}_j$ those discarded at overflow, $\mathcal{A}_j$ those accepted by at least one AUV, and $\mathcal{R}_j^{\mathrm{rel}}$ those released by valid ACKs. The stated acceptance and ACK rules imply

$$\mathcal{R}_j^{\mathrm{rel}} \subseteq \mathcal{A}_j \subseteq \mathcal{I}_j \setminus \mathcal{O}_j. \tag{S34}$$

These sets are disjoint across sensors because their keys include the sensor identifier. Consequently,

$$\begin{aligned} \sum_j R_j &= \sum_j |\mathcal{R}_j^{\mathrm{rel}}| \le \sum_j |\mathcal{A}_j| = C(T), \\ C(T) &= \sum_j |\mathcal{A}_j| \le \sum_j (|\mathcal{I}_j| - |\mathcal{O}_j|) \\ &= \sum_j \big(Q_{j,0} + N_j - O_j\big), \end{aligned} \tag{S35}$$

which establishes both inequalities. Retransmitting a key does not alter its membership or its cardinality. ∎

Corollary S.3 (Supply-side bound on expected native payload)**.** *For a full-horizon $T_{\max}$ with M independent or dependent Poisson-arrival sensors of common mean rate λ, and initial occupancy Q_0 per sensor,*

$$\mathbb{E}[C(T_{\max})] \le M(Q_0 + \lambda T_{\max}). \tag{S36}$$

Proof: Apply expectation to (S32). Because overflow is nonnegative and $\mathbb{E}[N_j(T_{\max})] = \lambda T_{\max}$,

$$\begin{aligned} \mathbb{E}[C(T_{\max})] &\le \sum_{j=1}^{M} \big(Q_0 + \mathbb{E}[N_j(T_{\max})] - \mathbb{E}[O_j(T_{\max})]\big) \\ &\le \sum_{j=1}^{M} (Q_0 + \lambda T_{\max}) \\ &= M(Q_0 + \lambda T_{\max}). \end{aligned} \tag{S37}$$

Linearity of expectation is sufficient; the sensor arrival processes need not be mutually independent. ∎

For the native 30-sensor setup, the resulting expectation bound is 192 packets; TDMA scheduling and collection mobility impose additional limits. The distinction between credited DATA and ACK-confirmed buffer release also carries forward to the resource-allocation analysis.

C. Exclusive Resource Ownership under Nested Leases

Conservation constrains the application data ledger; the same packet-level execution requires an independent invariant for ownership of scheduled acoustic opportunities.

Let $\mathcal{W}_v^{\mathrm{home}}$ be the fixed calendar of node v, $\mathcal{L}_v$ its locally installed authorization certificates, and $\mathcal{P}_v$ its irrevocable-until-expiration silence commitments. Consistent with main Eq. (34), define

$$\mathcal{W}_v^{\mathrm{use}}(t) = \left(\mathcal{W}_v^{\mathrm{home}} \cup \bigcup_{\lambda \in \mathcal{L}_v(t)} \mathcal{W}_\lambda \right) \setminus \bigcup_{\pi \in \mathcal{P}_v(t)} \mathcal{W}_\pi. \quad \text{(S38)}$$

The model assumes unique initial home-slot ownership, globally interpretable slot indices, valid nonforgeable certificates, and protocol-compliant nodes that never reclaim a promised window before its fixed expiration. A parent may delegate only windows that it owns and has not already promised to another child.

Proposition S.4 (Hereditary single-owner access)**.** *Suppose every successful delegation records the donor's silence before the borrower is permitted to activate rights, and each child certificate is a subset of an unexpired parent window that is not simultaneously delegated to another child. Then any finite chain of parent-to-child sub-delegations preserves*

$$\sum_{v \in \mathcal{N}} \mathbf{1}\{s \in \mathcal{W}_v^{\mathrm{use}}(t)\} \le 1 \quad \text{(S39)}$$

for every slot s and time t, under arbitrary packet loss, duplication, reordering, and finite delay that respect certificate expiration and the modeled event order.

Proof: Fix a calendar slot s and let $O_s(t) = \sum_v \ell_{v,s}(t)$ be its authorized-owner count. Initially, unique home ownership gives $O_s(0) \le 1$. We examine the transitions that can change this count. When a donor d accepts PREPARE, it records silence before any recipient is authorized. Therefore,

$$\begin{aligned} &\ell_{d,s}(t^+) \le \ell_{d,s}(t^-), \\ &\ell_{v,s}(t^+) = \ell_{v,s}(t^-), \quad v \ne d, \\ &O_s(t^+) \le O_s(t^-) \le 1. \end{aligned} \quad \text{(S40)}$$

A transmitted acknowledgment or queued COMMIT leaves all borrower rights unchanged. When a valid COMMIT is accepted by borrower b, the donor's overlapping right has already been removed. Because the donor was the unique predecessor owner and all sibling grants are nonoverlapping, the activation precondition is

$$\ell_{b,s}(t^+) = 1 \implies \sum_{v \ne b} \ell_{v,s}(t^+) = 0, \quad O_s(t^+) = 1. \quad \text{(S41)}$$

For nested delegation, the same implication follows inductively. If a parent owns window $\mathcal{W}_p$ and children receive windows $\mathcal{W}_{c_j}$, validity requires

$$\begin{aligned} \mathcal{W}_{c_j} &\subseteq \mathcal{W}_p, \\ \mathcal{W}_{c_j} \cap \mathcal{W}_{c_k} &= \varnothing, \qquad j \ne k, \end{aligned} \quad \text{(S42)}$$

with the parent's silence recorded before each activation. Thus extending a valid chain by one child preserves $O_s(t) \le 1$. Duplicate certificates do not create distinct rights. At a common window expiry the child's certificate expires together with its corresponding parent promise; any restored home right cannot coexist with a still-valid child right. Induction over event order and delegation depth therefore yields $O_s(t) \le 1$ at every admissible event time and, by the interval-validity rule, between events. ∎

The invariant establishes exclusive authorization under synchronized, protocol-compliant slot operation. Actual packet reception additionally depends on propagation, channel quality, and route availability, which motivates the following multihop analysis.

D. Conditional Completion on Installed Multihop Routes

Following exclusive resource authorization, multihop completion further depends on installed route versions, timely slots, and successful physical reception.

A requested rooted topology is an intention; an operational route must be assembled from receiver-local installed forwarding records and a succession of feasible physical hop events. For a packet with route version ν and path $v_h \to v_{h-1} \to \cdots \to v_0$, define $I_{v_j}^{\nu}(t)$ as the valid installed-version indicator at the forwarding instant and $\mathcal{B}_j$ as successful, timely reception on hop j after its scheduled attempt. A route path is operational through the deadline only if all required installed records persist, the packet remains valid, the necessary opportunities occur, and each scheduled hop is accepted.

Proposition S.5 (Conditional multihop completion bound)**.** *Fix a packet and an event window $[t_0, t_0 + D]$. Suppose (i) the packet has a valid route-version chain for every required forwarder throughout its relevant attempts; (ii) each next-hop transmission is offered an authorized opportunity early enough that all $h \le 2$ attempts can complete by the deadline if prior hops succeed; and (iii) on the history conditioned upon all previous successful hops, the next hop succeeds with probability at least $q_j > 0$. Then*

$$\mathbb{P}\{\textit{packet accepted at } v_0 \textit{ by } t_0 + D\} \ge \prod_{j=1}^{h} q_j. \quad \text{(S43)}$$

Proof: Let O denote the stipulated operational-route, authorization, and deadline conditions for the analyzed attempt sequence. Let $\mathcal{B}_j$ denote timely acceptance at hop j, conditional on all preceding hops being delivered. Under O, success of all h hops implies final acceptance event $\mathcal{D}_0$:

$$\mathsf{O} \cap \bigcap_{j=1}^{h} \mathcal{B}_j \subseteq \mathcal{D}_0. \quad \text{(S44)}$$

The probability chain rule gives, without hop independence,

$$\begin{aligned} \mathbb{P}(\mathcal{D}_0 \mid \mathsf{O}) &\ge \mathbb{P}\left(\bigcap_{j=1}^{h} \mathcal{B}_j \mid \mathsf{O} \right) \\ &= \prod_{j=1}^{h} \mathbb{P}(\mathcal{B}_j \mid \mathsf{O}, \mathcal{B}_1, \ldots, \mathcal{B}_{j-1}) \\ &\ge \prod_{j=1}^{h} q_j. \end{aligned} \quad \text{(S45)}$$

The proposition assumes O throughout the fixed execution window, so its unconditional bound follows. If the operational premises hold only on an event of nonunit probability, total probability instead yields

$$\begin{aligned}\mathbb{P}(\mathcal{D}_0) &\geq \mathbb{P}(\mathsf{O} \cap \mathcal{D}_0) = \mathbb{P}(\mathsf{O})\mathbb{P}(\mathcal{D}_0 \mid \mathsf{O}) \\ &\geq \mathbb{P}(\mathsf{O}) \prod_{j=1}^{h} q_j, \end{aligned} \tag{S46}$$

provided the same conditional hop bounds hold on O. Neither statement assigns a positive probability to a physically disconnected path. ∎

The lower bound applies when the path remains operational and its hop-level conditions hold. It therefore separates two essential mechanisms: configuration execution establishes an available forwarding path, and link delivery determines whether scheduled attempts succeed. This distinction leads naturally to the information-freshness analysis.

Building on the causal delivery and authorization properties, we next analyze the freshness and estimation consequences of the resulting packet arrivals. These results also characterize how maintaining usable coordination information competes with application-data transmission opportunities.

E. Information-Age Dynamics and Conditional Renewal Bounds

Following the source-time definition of information age [1], for AUV i, let $\tau_i^{\uparrow}(t)$ be the largest source timestamp accepted at the USV by t. If no accepted STATE with a newer source timestamp reaches the USV during $[a, b]$, then $\tau_i^{\uparrow}$ is constant on that interval. It follows immediately that

$$A_i^{\uparrow}(b) = A_i^{\uparrow}(a) + (b - a). \tag{S47}$$

The same rule applies when transmissions continue but every potentially fresher update is dropped, rejected, or rendered unusable by a missing installed path. The identity explains why reconfiguration requests that do not establish a working route cannot by themselves reduce USV-side STATE AoI. Conversely, an AUV may continue collecting nearby sensor DATA during a root partition because its task-data endpoint is local rather than the USV.

Divide an active interval into blocks of duration T_b. For finite $D_{\mathrm{fresh}} \geq 0$, let $\mathsf{E}_{i,n}$ be the event that, by the end of block n, the USV has accepted an AUV-i STATE update sourced no earlier than $nT_b - D_{\mathrm{fresh}}$. Suppose for every relevant prior history $\mathscr{F}_{n-1}$ that

$$\mathbb{P}(\mathsf{E}_{i,n} \mid \mathscr{F}_{n-1}) \geq q_i > 0. \tag{S48}$$

Here q_i represents a lower bound on end-to-end valid updates, incorporating access, queueing, route installation, and reception. In the same manner as the main theorem, for $1 \leq m \leq n$,

$$\mathbb{P}\{A_i^{\uparrow}(nT_b) > D_{\mathrm{fresh}} + mT_b\} \leq (1 - q_i)^m, \tag{S49}$$

$$\begin{aligned}\mathbb{E}[A_i^{\uparrow}(nT_b)] \leq\ & D_{\mathrm{fresh}} + (1 - q_i)^n Y_{i,0} \\ & + \frac{T_b(1 - q_i)}{q_i}[1 - (1 - q_i)^n]. \end{aligned} \tag{S50}$$

where $Y_{i,0} = [A_i^{\uparrow}(0) - D_{\mathrm{fresh}}]^+$. To expose the intermediate renewal argument, observe that a successful fresh update in any of the last m blocks prevents an age excursion beyond the stated threshold:

$$\begin{aligned}&\{A_i^{\uparrow}(nT_b) > D_{\mathrm{fresh}} + mT_b\} \\ &\quad \subseteq \mathsf{H}_{i,n,m}, \qquad \mathsf{H}_{i,n,m} = \bigcap_{r=n-m+1}^{n} \mathsf{E}_{i,r}^{c}. \end{aligned} \tag{S51}$$

Taking the intersection for $m = 0$ to be the sure event, conditioning on the preceding block history gives the geometric tail bound without an independence assumption:

$$\begin{aligned}\mathbb{P}(\mathsf{H}_{i,n,m}) &= \mathbb{E}\big[\mathbf{1}_{\mathsf{H}_{i,n-1,m-1}} \mathbb{P}(\mathsf{E}_{i,n}^{c} \mid \mathscr{F}_{n-1})\big] \\ &\leq (1 - q_i)\mathbb{P}(\mathsf{H}_{i,n-1,m-1}) \\ &\leq (1 - q_i)^m. \end{aligned} \tag{S52}$$

For the first moment, set $Y_{i,n} = [A_i^{\uparrow}(nT_b) - D_{\mathrm{fresh}}]^+$. A qualifying update resets the excess age; otherwise the excess age grows by at most T_b:

$$\begin{aligned}Y_{i,n} &\leq \mathbf{1}_{\mathsf{E}_{i,n}^{c}}(Y_{i,n-1} + T_b), \\ \mathbb{E}[Y_{i,n} \mid \mathscr{F}_{n-1}] &\leq (1 - q_i)(Y_{i,n-1} + T_b), \end{aligned} \tag{S53}$$

$$\begin{aligned}\mathbb{E}[Y_{i,n}] &\leq (1 - q_i)^n Y_{i,0} + T_b(1 - q_i) \sum_{r=0}^{n-1} (1 - q_i)^r \\ &= (1 - q_i)^n Y_{i,0} + \frac{T_b(1 - q_i)}{q_i}[1 - (1 - q_i)^n]. \end{aligned} \tag{S54}$$

Because $A_i^{\uparrow}(nT_b) \leq D_{\mathrm{fresh}} + Y_{i,n}$, these steps recover (S50) at active block boundaries. We next extend this renewal result to the frequency of stale-information events.

Corollary S.6 (Fraction of stale block boundaries)**.** *For any $m \geq 1$ and $n \geq m$, define the random fraction of post-warm-up block boundaries exceeding threshold $D_{\mathrm{fresh}} + mT_b$ as*

$$Z_{i,m,n} = \frac{1}{n - m + 1} \sum_{r=m}^{n} \mathbf{1}\{A_i^{\uparrow}(rT_b) > D_{\mathrm{fresh}} + mT_b\}. \tag{S55}$$

Under (S48)*,*

$$\mathbb{E}[Z_{i,m,n}] \leq (1 - q_i)^m. \tag{S56}$$

Furthermore, let $H_{i,m:n}$ denote the event that at least one monitored block-boundary age exceeds the same threshold. Then

$$\mathbb{P}(H_{i,m:n}) \leq (n - m + 1)(1 - q_i)^m. \tag{S57}$$

Proof: Introduce $H_{i,r,m} = \{A_i^{\uparrow}(rT_b) > D_{\mathrm{fresh}} + mT_b\}$. From (S49), $\mathbb{P}(H_{i,r,m}) \leq (1 - q_i)^m$ for $r \geq m$. Hence the expected fraction is

$$\begin{aligned}\mathbb{E}[Z_{i,m,n}] &= \frac{1}{n - m + 1} \sum_{r=m}^{n} \mathbb{E}[\mathbf{1}_{H_{i,r,m}}] \\ &= \frac{1}{n - m + 1} \sum_{r=m}^{n} \mathbb{P}(H_{i,r,m}) \\ &\leq (1 - q_i)^m. \end{aligned} \tag{S58}$$

For the maximum-age event, write $H_{i,m:n} = \bigcup_{r=m}^{n} H_{i,r,m}$. Boole's inequality gives

$$\begin{aligned}\mathbb{P}(H_{i,m:n}) &= \mathbb{P}\left(\bigcup_{r=m}^{n} H_{i,r,m}\right)\\ &\le \sum_{r=m}^{n} \mathbb{P}(H_{i,r,m})\\ &\le (n-m+1)(1-q_i)^m. \end{aligned} \tag{S59}$$

Here $H_{i,m:n}$ is precisely the event that the maximum monitored block-boundary age exceeds the threshold. No temporal independence is used in either step. ∎

The fraction and maximum-event bounds complement mean AoI by characterizing prolonged staleness. During periods without valid refreshes, including root partitions, the deterministic growth relation (S47) describes the resulting age evolution.

F. Disturbance-Filter Error and Motion-Prediction Bounds

Freshness controls the age of received evidence; the accuracy of its extrapolation additionally depends on disturbance estimation and motion variation. We analyze those effects next.

The USDE update in (S17) filters the measured velocity residual. Its tracking error exposes the tradeoff between suppressing measurement noise and responding to changing currents. For a sequence of valid displacement intervals of equal duration h, write

$$\boldsymbol{z}_k = \overline{\boldsymbol{d}}_k + \boldsymbol{\nu}_k, \quad \eta = 1 - e^{-h/\kappa}, \quad \widehat{\boldsymbol{d}}_k = (1-\eta)\widehat{\boldsymbol{d}}_{k-1} + \eta \boldsymbol{z}_k, \tag{S60}$$

where $\overline{\boldsymbol{d}}_k$ is the interval-averaged true current and $\boldsymbol{\nu}_k$ is the measured displacement error divided by h. This uses the same command integral as the actual estimator; delayed localization corrections are not part of $\boldsymbol{\nu}_k$.

Proposition S.7 (Tracking error of the residual smoother)**.** *Suppose consecutive valid intervals satisfy $\|\overline{\boldsymbol{d}}_k - \overline{\boldsymbol{d}}_{k-1}\| \le \Delta_d$ and $\|\boldsymbol{\nu}_k\| \le \epsilon_z$. If $\boldsymbol{e}_k = \widehat{\boldsymbol{d}}_k - \overline{\boldsymbol{d}}_k$, then*

$$\begin{aligned}\|\boldsymbol{e}_k\| &\le (1-\eta)^k \|\boldsymbol{e}_0\|\\ &\quad + \left[\epsilon_z + \frac{1-\eta}{\eta}\Delta_d\right]\left[1-(1-\eta)^k\right]. \end{aligned} \tag{S61}$$

If the current is constant over these intervals and independent zero-mean residual noise has common covariance Σ_ν, the limiting estimation-error covariance, when it exists, is

$$\lim_{k\to\infty} \mathrm{Cov}(\boldsymbol{e}_k) = \frac{\eta}{2-\eta}\Sigma_\nu. \tag{S62}$$

Proof: Set $\alpha = 1-\eta \in (0,1)$. Subtracting the true interval-average disturbance from (S60) gives

$$\boldsymbol{e}_k = \alpha \boldsymbol{e}_{k-1} + \alpha(\overline{\boldsymbol{d}}_{k-1} - \overline{\boldsymbol{d}}_k) + \eta \boldsymbol{\nu}_k, \tag{S63}$$

$$\|\boldsymbol{e}_k\| \le \alpha \|\boldsymbol{e}_{k-1}\| + \alpha\Delta_d + \eta\epsilon_z. \tag{S64}$$

Repeated substitution yields the finite geometric expansion

$$\begin{aligned}\|\boldsymbol{e}_k\| &\le \alpha^k \|\boldsymbol{e}_0\| + (\alpha\Delta_d + \eta\epsilon_z)\sum_{r=0}^{k-1}\alpha^r\\ &= \alpha^k\|\boldsymbol{e}_0\| + \frac{\alpha\Delta_d + \eta\epsilon_z}{1-\alpha}(1-\alpha^k)\\ &= \alpha^k\|\boldsymbol{e}_0\| + \left(\epsilon_z + \frac{1-\eta}{\eta}\Delta_d\right)(1-\alpha^k), \end{aligned} \tag{S65}$$

which proves the norm bound. For constant $\overline{\boldsymbol{d}}_k$, assume that fresh zero-mean residual noise is independent of the prior estimation error and has common covariance Σ_ν. Then

$$\begin{aligned}\Sigma_{e,k} &= \alpha^2\Sigma_{e,k-1} + \eta^2\Sigma_\nu\\ &= \alpha^{2k}\Sigma_{e,0} + \eta^2\sum_{r=0}^{k-1}\alpha^{2r}\Sigma_\nu\\ &= \alpha^{2k}\Sigma_{e,0} + \frac{\eta^2(1-\alpha^{2k})}{1-\alpha^2}\Sigma_\nu. \end{aligned} \tag{S66}$$

Since $1-\alpha^2 = \eta(2-\eta)$ and $\alpha^{2k} \to 0$,

$$\lim_{k\to\infty}\Sigma_{e,k} = \frac{\eta}{2-\eta}\Sigma_\nu. \tag{S67}$$

The bounded-noise and stationary-noise claims use separate assumptions; neither asserts a deterministic bound for an untruncated Gaussian residual. ∎

The bound quantifies the smoothing–adaptation tradeoff: smaller η suppresses stationary residual noise but increases sensitivity to current variation. The deterministic bound applies on bounded-error events; during sensing dropouts, the filter retains its last estimate and tracks the age of that evidence. These estimation effects propagate into the remote motion predictor considered next.

The main paper uses the USV's damped persistence predictor

$$\begin{aligned}\widetilde{\boldsymbol{p}}_i(t+H) &= \widehat{\boldsymbol{p}}_i(\tau_i) + g(s_i)\widehat{\boldsymbol{v}}_i,\\ s_i &= A_i^{\uparrow}(t) + H, \quad g(s) = \tau_v(1-e^{-s/\tau_v}). \end{aligned} \tag{S68}$$

with $\widehat{\boldsymbol{v}}_i$ derived from the last delivered command and initialized disturbance estimate. Let $\varepsilon_{p,i}$ and $\varepsilon_{v,i}$ upper-bound the source-time position and velocity estimation errors on a given event, and let V_i upper-bound true ground speed on the analyzed interval. The main proposition gives

$$\begin{aligned}&\|\boldsymbol{p}_i(t+H) - \widetilde{\boldsymbol{p}}_i(t+H)\|\\ &\quad \le \varepsilon_{p,i} + g(s_i)\varepsilon_{v,i} + [s_i - g(s_i)]V_i + \mathcal{D}_i(s_i). \end{aligned} \tag{S69}$$

where $\mathcal{D}_i(s) = \int_0^s \|\boldsymbol{v}_i^g(\tau_i + r) - \boldsymbol{v}_i^g(\tau_i)\| dr$. The final term is important because a learned AUV can change commands within the slow USV horizon even with a perfect current estimate.

Corollary S.8 (Bounded velocity-variation refinement)**.** *If, in addition to the source-time error bounds, true ground velocity satisfies $\|\boldsymbol{v}_i^g(\tau_i + r) - \boldsymbol{v}_i^g(\tau_i)\| \le a_i r$ over $r \in [0, s_i]$, then*

$$\begin{aligned}&\|\boldsymbol{p}_i(t+H) - \widetilde{\boldsymbol{p}}_i(t+H)\| \le \mathcal{E}_i(s_i),\\ &\mathcal{E}_i(s) = \varepsilon_{p,i} + g(s)\varepsilon_{v,i}\\ &\qquad + [s - g(s)]V_i + \tfrac{1}{2}a_i s^2. \end{aligned} \tag{S70}$$

Proof: Write $\boldsymbol{v}_i^g = \dot{\boldsymbol{p}}_i$ and $\boldsymbol{\delta}_{p,i} = \boldsymbol{p}_i(\tau_i) - \widehat{\boldsymbol{p}}_i(\tau_i)$, $\boldsymbol{\delta}_{v,i} = \boldsymbol{v}_i^g(\tau_i) - \widehat{\boldsymbol{v}}_i$. Integrating the true motion and subtracting the predictor in (S68), with $s = s_i$, gives

$$\begin{aligned}\boldsymbol{p}_i(\tau_i + s) - \widetilde{\boldsymbol{p}}_i(\tau_i + s) = \boldsymbol{\delta}_{p,i} + g(s)\boldsymbol{\delta}_{v,i} \\ + [s - g(s)]\boldsymbol{v}_i^g(\tau_i) \\ + \int_0^s \big[\boldsymbol{v}_i^g(\tau_i + r) - \boldsymbol{v}_i^g(\tau_i)\big] dr.\end{aligned} \tag{S71}$$

The additional velocity-variation assumption implies

$$\mathcal{D}_i(s) \le \int_0^s a_i r \, dr = \frac{a_i s^2}{2}. \tag{S72}$$

Applying the triangle inequality to (S71), using $\|\boldsymbol{\delta}_{p,i}\| \le \varepsilon_{p,i}$, $\|\boldsymbol{\delta}_{v,i}\| \le \varepsilon_{v,i}$, and $\|\boldsymbol{v}_i^g(\tau_i)\| \le V_i$, yields

$$\begin{aligned}\|\boldsymbol{p}_i - \widetilde{\boldsymbol{p}}_i\| &\le \varepsilon_{p,i} + g(s)\varepsilon_{v,i} + (s - g(s))V_i + \mathcal{D}_i(s) \\ &\le \varepsilon_{p,i} + g(s)\varepsilon_{v,i} + (s - g(s))V_i + \tfrac{1}{2}a_i s^2.\end{aligned} \tag{S73}$$

Putting $s = s_i$ proves the stated refinement. ∎

The refinement applies when velocity variation is bounded over the prediction window; abrupt command switches can instead be handled through the original variation term. The source-time error constants are interpreted under the corresponding bounded-error conditions. We now examine how this endpoint uncertainty affects the USV positioning score.

G. FIM Sensitivity and Age-Dependent Position-Selection Risk

The forecast enters the USV planner through a finite set of geometric candidates. The next bounds quantify how endpoint uncertainty can perturb the FIM-based score, subject to a fixed planning snapshot.

We now spell out the geometric step used by the main-paper positioning proposition. For horizontal relative displacement $\boldsymbol{z} = \boldsymbol{p}_i - \boldsymbol{x}$, let $D = (\|\boldsymbol{z}\|^2 + h_A^2)^{1/2}$ and $\boldsymbol{e} = \boldsymbol{z}/D$. The implemented horizontal range–direction FIM can be written

$$\boldsymbol{F}(\boldsymbol{z}) = \frac{\boldsymbol{e}\boldsymbol{e}^{\mathsf{T}}}{\sigma_r^2} + \frac{\boldsymbol{I}_2 - \boldsymbol{e}\boldsymbol{e}^{\mathsf{T}}}{\sigma_b^2 D^2}. \tag{S74}$$

Because $D \ge h_A > 0$ and $\|\boldsymbol{e}\| \le 1$, the directional derivative in a horizontal increment $\boldsymbol{u}$ is

$$\begin{aligned}\mathrm{D}\boldsymbol{e}(\boldsymbol{z})[\boldsymbol{u}] &= \frac{\boldsymbol{u}}{D} - \frac{\boldsymbol{z}(\boldsymbol{z}^{\mathsf{T}}\boldsymbol{u})}{D^3}, \\ \|\mathrm{D}\boldsymbol{e}(\boldsymbol{z})\|_2 &\le D^{-1} \le h_A^{-1}.\end{aligned} \tag{S75}$$

Applying the product rule separately to the two terms in (S74) yields

$$\begin{aligned}\|\mathrm{D}(\boldsymbol{e}\boldsymbol{e}^{\mathsf{T}})\|_2 &\le 2h_A^{-1}, \\ \|\nabla D^{-2}\| &\le 2h_A^{-3}, \\ \|\mathrm{D}[(\boldsymbol{I}_2 - \boldsymbol{e}\boldsymbol{e}^{\mathsf{T}})D^{-2}]\|_2 &\le 4h_A^{-3}.\end{aligned} \tag{S76}$$

The triangle inequality and integration along the line segment from $\boldsymbol{z}$ to $\boldsymbol{z}'$ therefore give a global horizontal-position Lipschitz constant

$$L_F = \frac{2}{\sigma_r^2 h_A} + \frac{4}{\sigma_b^2 h_A^3}, \qquad \|\boldsymbol{F}(\boldsymbol{z}) - \boldsymbol{F}(\boldsymbol{z}')\|_2 \le L_F \|\boldsymbol{z} - \boldsymbol{z}'\|. \tag{S77}$$

For positive-semidefinite matrices $\boldsymbol{F}, \boldsymbol{F}'$ and $c \ge 0$, define $\boldsymbol{F}_\alpha = (1-\alpha)\boldsymbol{F} + \alpha\boldsymbol{F}'$. Since $\boldsymbol{I}_2 + c\boldsymbol{F}_\alpha \succeq \boldsymbol{I}_2$,

$$\begin{aligned}&|\log\det(\boldsymbol{I}_2 + c\boldsymbol{F}') - \log\det(\boldsymbol{I}_2 + c\boldsymbol{F})| \\ &\quad = \left| \int_0^1 c \, \mathrm{tr}\big[(\boldsymbol{I}_2 + c\boldsymbol{F}_\alpha)^{-1}(\boldsymbol{F}' - \boldsymbol{F})\big] \, d\alpha \right| \\ &\quad \le 2c\|\boldsymbol{F}' - \boldsymbol{F}\|_2.\end{aligned} \tag{S78}$$

Indeed, the two-dimensional trace bound used in the last step follows from

$$\begin{aligned}&\left|\mathrm{tr}\left[(\boldsymbol{I}_2 + c\boldsymbol{F}_\alpha)^{-1}(\boldsymbol{F}' - \boldsymbol{F})\right]\right| \\ &\quad \le 2\|(\boldsymbol{I}_2 + c\boldsymbol{F}_\alpha)^{-1}\|_2 \|\boldsymbol{F}' - \boldsymbol{F}\|_2 \\ &\quad \le 2\|\boldsymbol{F}' - \boldsymbol{F}\|_2.\end{aligned} \tag{S79}$$

The direct-route distance term is separately 1-Lipschitz in the AUV endpoint, whereas the USV movement term is independent of it. Hence, with the fixed received-report weights and route state,

$$\begin{aligned}\sup_{\boldsymbol{x} \in \mathcal{P}_0(t)} |J_0(\boldsymbol{x}) - J_0^{\mathrm{ref}}(\boldsymbol{x})| &\le \sum_{i \in \mathcal{V}(t)} L_i E_i(s_i) \\ &= \Delta_J(t).\end{aligned} \tag{S80}$$

The resulting bound compares candidate scores at a fixed planning snapshot. Combining it with the AoI renewal bound yields a probabilistic characterization of candidate-selection sensitivity.

The next statement links the age result to the USV planning score under an explicit intersection of assumptions. Let $t = nT_b$, suppose the assumptions of Corollary S.8 hold uniformly on the event being studied with fixed nonnegative constants, and define $s_m = D_{\mathrm{fresh}} + mT_b + H$. Since $0 \le g(s) \le s$ and both $g(s)$ and $s - g(s)$ are nondecreasing, the envelope $\mathcal{E}_i(s)$ in (S70) is nondecreasing.

Proposition S.9 (Conditional age-to-score coupling)**.** *Suppose each valid received AUV state satisfies* (S48) *with its own $q_i > 0$, the error/variation bounds hold on the analyzed snapshot, and the USV candidate set, accepted report weights, and confirmed topology are held fixed for sensitivity comparison as in the main paper. Let L_i be the corresponding nonnegative position-score Lipschitz coefficient and set*

$$\Delta_m(t) = \sum_{i \in \mathcal{V}(t)} L_i \mathcal{E}_i(s_m). \tag{S81}$$

Define the geometric reference regret using true endpoints on the same realized trajectory:

$$\mathfrak{R}_0(t) = J_0^{\mathrm{ref}}(\boldsymbol{x}^\star) - \min_{\boldsymbol{x} \in \mathcal{P}_0(t)} J_0^{\mathrm{ref}}(\boldsymbol{x}). \tag{S82}$$

If the uniform estimation and motion-error conditions hold surely throughout the comparison, then

$$\mathbb{P}\{\mathfrak{R}_0(t) > 2\Delta_m(t)\} \le \sum_{i=1}^{N} (1 - q_i)^m. \tag{S83}$$

If those conditions or fixed-snapshot admissibility assumptions fail with probability ϵ_{aux}, add ϵ_{aux} to the right-hand side; without such a bound the claim is conditional only.

Proof: Let $t = nT_b$ and define the joint freshness event

$$\mathsf{G}_m = \bigcap_{i=1}^{N} \{A_i^{\uparrow}(t) \le D_{\mathrm{fresh}} + mT_b\}. \tag{S84}$$

On G_m, every usable report satisfies $s_i = A_i^{\uparrow}(t) + H \le s_m$. The nondecreasing bound (S70) then gives

$$\begin{aligned} &\|\boldsymbol{p}_i(t+H) - \widetilde{\boldsymbol{p}}_i(t+H)\| \le \mathcal{E}_i(s_i) \le \mathcal{E}_i(s_m), \\ &\sup_{\boldsymbol{x}\in\mathcal{P}_0(t)} |J_0(\boldsymbol{x}) - J_0^{\mathrm{ref}}(\boldsymbol{x})| \le \sum_{i\in\mathcal{V}(t)} L_i \mathcal{E}_i(s_m) = \Delta_m(t). \end{aligned} \tag{S85}$$

Let $\boldsymbol{x}^{\dagger}$ minimize J_0^{ref} and $\boldsymbol{x}^{\star}$ minimize J_0 on the common fixed candidate set. Their optimality and the preceding uniform deviation imply

$$\begin{aligned} \mathfrak{R}_0(t) &= J_0^{\mathrm{ref}}(\boldsymbol{x}^{\star}) - J_0^{\mathrm{ref}}(\boldsymbol{x}^{\dagger}) \\ &\le J_0(\boldsymbol{x}^{\star}) + \Delta_m - J_0^{\mathrm{ref}}(\boldsymbol{x}^{\dagger}) \\ &\le J_0(\boldsymbol{x}^{\dagger}) + \Delta_m - J_0^{\mathrm{ref}}(\boldsymbol{x}^{\dagger}) \\ &\le 2\Delta_m(t). \end{aligned} \tag{S86}$$

Let C be the event on which all stated motion-error and fixed-snapshot conditions hold, and define $\mathsf{R}_m = \{\mathfrak{R}_0(t) > 2\Delta_m(t)\}$. The preceding pathwise implication and the union bound give

$$\begin{aligned} \mathsf{R}_m &\subseteq \mathsf{G}_m^c \cup \mathsf{C}^c, \\ \mathbb{P}(\mathsf{R}_m) &\le \mathbb{P}(\mathsf{G}_m^c) + \mathbb{P}(\mathsf{C}^c) \\ &\le \sum_{i=1}^{N} (1-q_i)^m + \epsilon_{\mathrm{aux}}. \end{aligned} \tag{S87}$$

Taking $\mathbb{P}(\mathsf{C}^c) = 0$ recovers the stated bound. The union bound does not require independent deliveries across AUVs. ∎

The result connects timely STATE reception with the stability of a finite-candidate geometric decision under the stated motion and estimation conditions. It also motivates a deterministic margin criterion for identifying when the selected USV target remains unchanged.

Corollary S.10 (Stability of the selected USV candidate)**.** *At a fixed snapshot, suppose the reference score has a unique minimizer $\boldsymbol{x}^{\dagger}$ over the finite candidate set and its gap to every alternative exceeds $2\Delta_J$, where Δ_J uniformly bounds $|J_0 - J_0^{\mathrm{ref}}|$. Then the implemented minimizer of J_0 equals $\boldsymbol{x}^{\dagger}$.*

Proof: For any candidate $\boldsymbol{x} \neq \boldsymbol{x}^{\dagger}$, apply the two-sided uniform score bound to obtain

$$\begin{aligned} J_0(\boldsymbol{x}) - J_0(\boldsymbol{x}^{\dagger}) &\ge J_0^{\mathrm{ref}}(\boldsymbol{x}) - \Delta_J - J_0^{\mathrm{ref}}(\boldsymbol{x}^{\dagger}) - \Delta_J \\ &= \left[J_0^{\mathrm{ref}}(\boldsymbol{x}) - J_0^{\mathrm{ref}}(\boldsymbol{x}^{\dagger})\right] - 2\Delta_J \\ &> 0. \end{aligned} \tag{S88}$$

Hence $\boldsymbol{x}^{\dagger}$ also strictly minimizes the implemented score J_0, establishing equality of the selected candidates. ∎

Thus a sufficiently large reference-score gap preserves the selected USV candidate despite bounded prediction error. When the gap is small, coordination decisions become more sensitive to information age and estimation quality.

H. Airtime Ledger, Coordination Benefits, and Return Consistency

A better location estimate or route may increase the chance of timely STATE delivery, yet each coordination packet also consumes a finite slot. We next separate resource-accounting effects from the learning objective across fast and slow decision clocks.

A TDMA scheduler has a fixed number of slots over a specified aligned horizon. For an evaluation window beginning with no in-flight DATA from earlier slots, classify its S_T scheduled slots as (i) slots emitting a non-DATA control/coordination packet, (ii) unused slots (including committed silence), or (iii) slots emitting a DATA packet. Let $N_{\mathrm{c}}, N_{\mathrm{idle}}, N_{\mathrm{DATA}}$ be their counts. Of the DATA emissions, let N_{unique} yield a first accepted sensor payload by the window deadline, and let N_{nc} fail to yield new credited payload by that deadline (failed, duplicate, rejected, or still in flight). By construction,

$$S_T = N_{\mathrm{c}} + N_{\mathrm{idle}} + N_{\mathrm{DATA}}, \tag{S89}$$

$$N_{\mathrm{DATA}} = N_{\mathrm{unique}} + N_{\mathrm{nc}}, \tag{S90}$$

$$\frac{B_{\mathrm{del}}(T)}{B_p} = S_T - N_{\mathrm{c}} - N_{\mathrm{idle}} - N_{\mathrm{nc}}. \tag{S91}$$

For fixed-budget accounting, scheduled slots after early episode termination are assigned to the unused/terminated category. An unfinished final physical slot must be excluded or separately classified so that S_T covers complete, nonoverlapping slots.

Proposition S.11 (Difference decomposition for two resource policies)**.** *Consider policies A and B with the same fixed total slot budget S_T, identical payload size B_p, and the preceding mutually exclusive ledger categories. Their distinct credited DATA difference obeys*

$$\frac{B_{\mathrm{del}}^A - B_{\mathrm{del}}^B}{B_p} = \left(N_{\mathrm{c}}^B - N_{\mathrm{c}}^A\right) + \left(N_{\mathrm{idle}}^B - N_{\mathrm{idle}}^A\right) + \left(N_{\mathrm{nc}}^B - N_{\mathrm{nc}}^A\right). \tag{S92}$$

Proof: Apply the mutually exclusive slot classification to each policy $X \in \{A, B\}$:

$$\begin{aligned} \frac{B_{\mathrm{del}}^X}{B_p} &= N_{\mathrm{unique}}^X \\ &= N_{\mathrm{DATA}}^X - N_{\mathrm{nc}}^X \\ &= S_T - N_{\mathrm{c}}^X - N_{\mathrm{idle}}^X - N_{\mathrm{nc}}^X. \end{aligned} \tag{S93}$$

Subtract the identity for B from that for A:

$$\begin{aligned} \frac{B_{\mathrm{del}}^A - B_{\mathrm{del}}^B}{B_p} &= \left[S_T - N_{\mathrm{c}}^A - N_{\mathrm{idle}}^A - N_{\mathrm{nc}}^A\right] \\ &\quad - \left[S_T - N_{\mathrm{c}}^B - N_{\mathrm{idle}}^B - N_{\mathrm{nc}}^B\right] \\ &= \Delta N_{\mathrm{c}}^{B-A} + \Delta N_{\mathrm{idle}}^{B-A} + \Delta N_{\mathrm{nc}}^{B-A}, \end{aligned} \tag{S94}$$

where $\Delta N_x^{B-A} = N_x^B - N_x^A$. The common slot budget cancels exactly; no independence assumption or throughput approximation is involved. ∎

The decomposition shows how leasing can improve payload by reducing idle slots while introducing control-handshake and unsuccessful-attempt costs. Topology repair may likewise trade payload airtime for fresher coordination, consistent with

the main-paper interventions. Reconstructing these terms from complete per-slot records enables direct attribution of resource-use differences.

Write $\mathcal{I}_i^{\mathrm{path}}(t)$ for the indicator that the AUV has a currently valid installed path to the USV under modeled hard-link feasibility, and $\mathcal{D}_i(t)$ for receipt of a fresh STATE update. Operational path availability is a prerequisite for ordinary routed delivery but is not itself delivery. Thus, for a path-dependent packet, an appropriate event decomposition is

$$\mathbb{P}(\mathcal{D}_i(t)) = \mathbb{P}(\mathcal{I}_i^{\mathrm{path}}(t))\mathbb{P}(\mathcal{D}_i(t) \mid \mathcal{I}_i^{\mathrm{path}}(t)) \tag{S95}$$

when delivery implies a valid installed path and the timing of the path event is defined over the relevant forwarding interval. The first factor is primarily affected by configuration installation and routing; the second is affected by queueing, authorized access, and successful physical reception. This decomposition should be evaluated with precise interval definitions, since instantaneous availability at the packet's final arrival time is not generally equivalent to availability across all forwarding attempts.

A third, different effect is application DATA service at the sensor–AUV link, which need not use the routed path to the USV. Thus lower AoI can coincide with lower sensor payload if more capacity is spent on routed coordination; higher local DATA collection can likewise coexist with worse USV-side freshness under root partition. These effects motivate treating payload, AoI, authorization utilization, and installed-path availability as separate response variables rather than exchanging one for another as a proxy.

Let slow-decision interval ℓ begin at fast step k_ℓ, contain d_ℓ actually executed fast steps, and accumulate topology rewards r_k^R as

$$R_\ell = \sum_{j=0}^{d_\ell-1} \gamma^j r_{k_\ell+j}^R. \tag{S96}$$

The duration convention follows temporal abstraction [4]; the exact discount across the interval is γ^{d_ℓ}. For a partition of n executed steps with $k_{\ell+1} = k_\ell + d_\ell$, the full expansion is

$$\begin{aligned}\sum_\ell \gamma^{k_\ell} R_\ell &= \sum_\ell \gamma^{k_\ell} \sum_{j=0}^{d_\ell-1} \gamma^j r_{k_\ell+j}^R \\ &= \sum_\ell \sum_{j=0}^{d_\ell-1} \gamma^{k_\ell+j} r_{k_\ell+j}^R \\ &= \sum_{k=0}^{n-1} \gamma^k r_k^R. \end{aligned} \tag{S97}$$

Writing $G_{k_\ell} = \sum_{k=k_\ell}^{n-1} \gamma^{k-k_\ell} r_k^R$ for the continuation return also gives

$$\begin{aligned} G_{k_\ell} &= \sum_{j=0}^{d_\ell-1} \gamma^j r_{k_\ell+j}^R + \gamma^{d_\ell} \sum_{k=k_{\ell+1}}^{n-1} \gamma^{k-k_{\ell+1}} r_k^R \\ &= R_\ell + \gamma^{d_\ell} G_{k_{\ell+1}}. \end{aligned} \tag{S98}$$

The aggregation identity also accommodates physically shortened intervals through their realized durations. Time-limit truncations follow the stated replay rule, preserving the intended discounting across fast and slow decisions.

The motion reward is $r_k = U_k + w_P(\gamma\Phi_{k+1} - \Phi_k)$, where U_k is the normalized operational utility. Potential-based shaping [5] admits the explicit telescoping calculation

$$\begin{aligned} &\sum_{k=0}^{n-1} \gamma^k(\gamma\Phi_{k+1} - \Phi_k) \\ &\quad = \sum_{k=1}^{n} \gamma^k \Phi_k - \sum_{k=0}^{n-1} \gamma^k \Phi_k \\ &\quad = \gamma^n \Phi_n - \Phi_0. \end{aligned} \tag{S99}$$

It follows pathwise that

$$\sum_{k=0}^{n-1} \gamma^k r_k = \sum_{k=0}^{n-1} \gamma^k U_k + w_P(\gamma^n \Phi_n - \Phi_0). \tag{S100}$$

For physical termination with $\Phi_n = 0$ and fixed initialization, the shaping difference is constant; for time-limit truncation, the terminal potential remains in the return. The auxiliary actor loss and topology-specific penalties introduce additional training preferences beyond this telescoping identity.

I. Physical Safety and Validity Boundaries

We finally connect these communication-level results to the physical termination criteria used in evaluation.

At each physical substep, collisions and boundary exits are checked against true AUV positions. These checks define mission termination, while the authorization invariant and position-error bounds characterize communication and estimation instead of physical collision avoidance. The distinction is particularly relevant to the four-AUV results, where collision and boundary terminations are frequent. Incorporating an explicit safety filter or barrier-based controller is a natural extension of the present framework.

Collectively, the results establish the internal consistency of receiver-local decisions, versioned communication execution, and cross-timescale return aggregation. The conditional age and positioning bounds make explicit the link between successful information renewal and the sensitivity of subsequent USV decisions, while the airtime identities quantify the accompanying coordination cost.

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